# Outbreak investigation

**Kiesha Prem** (PhD)

Saw Swee Hock School of Public Health

# Today's class

Outbreak Investigation Practical Briefing

Outbreak Investigation Lecture by Dr Gao Qi

# Outbreak investigation groupwork

Outbreak Investigation Practical Briefing (today)
Week 10 Outbreak Investigation Practical session 1
Week 11 Outbreak Investigation Practical session 2

More details are provided on the Outbreak Investigation Question Paper.

# Outbreak investigation Your Task



Several hospitals have notified MOH about a cluster of patients experiencing symptoms such as vomiting, diarrhoea, and abdominal cramps. Preliminary investigations reveal that many of the individuals affected had recently attended a public health conference held at a convention centre in downtown Singapore.

In light of this, the Ministry has initiated an outbreak investigation and your team has been tasked with leading the effort. A field investigation unit has already gathered initial data from conference participants. **Your role is to analyse the information, assess whether the conference was the likely source of the outbreak, identify the possible cause, and recommend strategies for control and prevention.**

# SPH2002: Outbreak investigation group assignment

**Learning objectives**

By the end of this assignment, students will be able to:

1. Apply the steps of outbreak investigation
2. Analyze epidemiological data to identify the source of an outbreak
3. Communicate findings clearly in an evidence-based report
4. Design appropriate control and prevention measures

**Your task**

Several hospitals have notified the Ministry of Health about a cluster of patients experiencing symptoms such as vomiting, diarrhoea, and abdominal cramps. Preliminary investigations reveal that many of the individuals affected had recently attended a public health conference held at a convention centre in downtown Singapore.

In light of this, the Ministry has initiated an outbreak investigation and your team has been tasked with leading the effort. A field investigation unit has already gathered initial data from conference participants. Your role is to analyse the information, assess whether the conference was the likely source of the outbreak, identify the possible cause, and recommend strategies for control and prevention.

**Assignment schedule**

<u>Week 9: Outbreak Investigation Lecture</u>

- Briefing on the assignment
- Guest lecture on outbreak investigation principles and case studies in Singapore

# Outbreak investigation

WHO. Foodborne Disease Outbreaks: Guidelines for Investigation and Control

# Outbreak investigation Practical Session 1

NUS National University of Singapore | Saw Swee Hock School of Public Health

Date: 24 & 31 March 2026, Tuesday

Venue: **Level 8, Learning Room 1**

- Each group will be provided with a unique dataset
- You will receive a set of questions to answer (already available in week 10 folder)
- Work in your groups to analyse the data
- Compile and submit a group report
- Relevant reading materials will be provided
- There will be facilitators during the session to help you

Group members: Login to Canvas > SPH2002 > People > Groups

**Unique dataset for each group**

**Canvas > SPH2002 > Files > Week 10 > Outbreak datasets**

Name

- sph2002_obdataset_group_T1G1.csv
- sph2002_obdataset_group_T1G2.csv
- sph2002_obdataset_group_T1G3.csv
- sph2002_obdataset_group_T1G4.csv
- sph2002_obdataset_group_T1G5.csv
- sph2002_obdataset_group_T2G1.csv
- sph2002_obdataset_group_T2G2.csv
- sph2002_obdataset_group_T2G3.csv
- sph2002_obdataset_group_T2G4.csv
- sph2002_obdataset_group_T2G5.csv
- sph2002_obdataset_group_T3G1.csv
- sph2002_obdataset_group_T3G2.csv

# Outbreak investigation

## Datasets

| Variable name | Description | Coding |
|---|---|---|
| **group** | Outbreak group number | |
| **unique_id** | Unique identifier for each individual | |
| **sex** | Sex of the individual | Male, Female |
| **age** | Age in years | |
| **residency_status** | Country of residence | Singapore, Overseas |
| **participant_role** | Role of the individual at the conference | Presenter, Delegate |
| **ate_start** | Date and time when individual started eating at the conference | |
| **illness_start_time** | Date and time when symptoms first appeared | NA: Don't know |
| **illness_end_time** | Date and time when individual recovered | NA: Don't know |
| **case** | Person was a case of illness | 1: Case; 0: Non-case |
| **vomit** | Person experienced vomiting | 1: Yes<br>0: No<br>NA: Don't know<br>8: Don't know |
| **diarr** | Person experienced diarrhoea | |
| **abdopain** | Person experienced abdominal pain | |
| **fever** | Person experienced fever | |
| **nausea** | Person experienced nausea | |
| **appetite** | Person experienced loss of appetite | |
| **caesar salad** | Person ate caesar salad | 1: ate/drank the item during wedding reception<br>0: did not eat/drink the item<br>NA: Don't know<br>8: Don't know |
| **smoked_salmon** | Person ate smoked salmon | |
| **cheese_platter** | Person ate cheese platter | |
| **roast_chicken** | Person ate roast chicken | |
| **grilled_salmon** | Person ate grilled salmon | |
| **beef_tenderloin** | Person ate beef tenderloin | |
| **mashed_potatoes** | Person ate mashed potatoes | |
| **mini_quiches** | Person ate mini quiches | |
| **sparkling_water** | Person drank sparkling water | |
| **red_wine** | Person drank red wine | |
| **white_wine** | Person drank white wine | |
| **orange_juice** | Person drank orange juice | |
| **lemon_cake** | Person ate lemon cake | |
| **chocolate_fondue** | Person ate chocolate fondue | |
| **Tiramisu** | Person ate tiramisu | |
| **Strawberry compote** | Person ate strawberry compote | |

# Outbreak investigation

## Submitting your assignment

# Outbreak investigation Submission

Date: 31 March 2026, 11:59 pm Tuesday

- Only one submission per group is required. Please ensure all group members agree on the final versions before submitting.
- Label both files using your assigned tutorial and group number. For example: Tutorial 1, Group 1 should label their files as 'Report_T1G1' and 'Analyses_T1G1'
- Upload your files to the designated folders on Canvas: "Outbreak Investigation Submission -- Report" and "Outbreak Investigation Submission – Analyses Output"
- **Canvas > SPH2002 > Assignments**
- Late submissions will incur a penalty of 5% deduction per day. No extensions will be granted.
- Each group will receive a group mark that will count 20% towards your grade for this course. Please note that the final group mark is subjected to peer review.
- **Peer review** will be open from 31 Mar to 7 Apr. No extensions.

# Outbreak investigation Submission

Date: 31 March 2026, 11:59 pm Tuesday

Assignments

Outbreak investigation submission_Report
Due 31 Mar at 23:59

Outbreak investigation submission_ Analyses Output
Due 31 Mar at 23:59

# Outbreak investigation Practical Sessions

Date: 24 & 31 March 2026, Tuesday

Venue: Level 8, Learning Room 1

- The sessions are designed for your group to **discuss and work on the assignment.**
- The facilitator is available to clarify any questions you may have, but **will not provide direct answers.**
- All necessary epidemiological concepts have been taught, and you are provided with references to help you.
- **Attendance is compulsory unless you have a valid reason.**
- Those absent should contact their group to follow up on the assigned tasks.

# Thank you!

kiesha.prem@nus.edu.sg